

CS-375 Assignment 4 Report:

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- B.1 Graph - Done
- B.2 Linear Programming - Done
- B.3 Simplex (Extra Credit) - Done

[Part B]: Programming Part (26%)

B.1 Graph Algorithm (16%)

Suppose a CS curriculum consists of n courses, all of them mandatory. The prerequisite graph G has a node for each course, and an edge from course v to course w if and only if v is a prerequisite for w . Find an algorithm that works directly with this graph representation, and computes the minimum number of semesters necessary to complete the curriculum (Assume that a student can take any number of courses in one semester).

Using following example to justify your answer:

The CS Department requires fifteen one-semester courses with the prerequisites shown below:

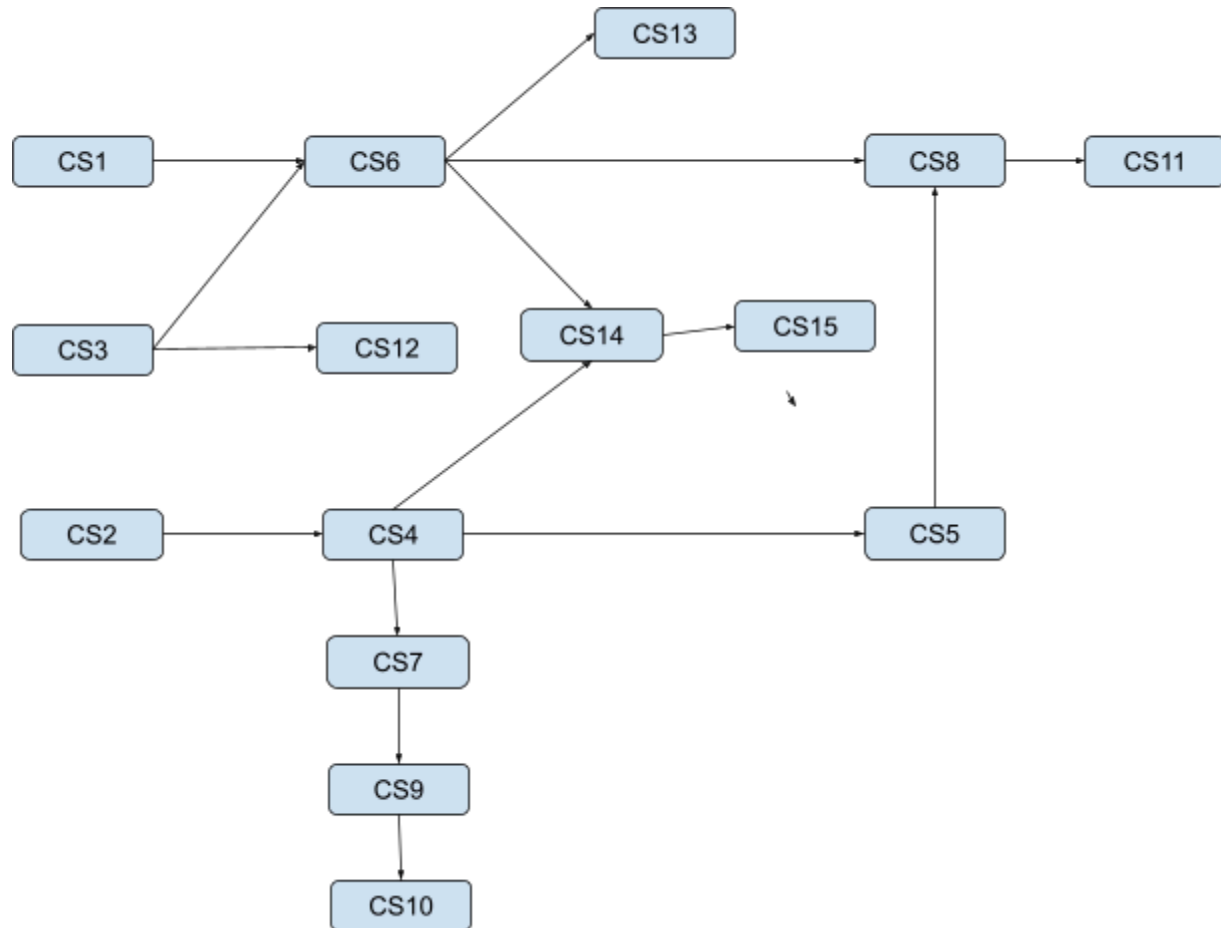
cs1
cs2
cs3
cs4 requires cs2
cs5 requires cs4
cs6 requires cs1 and cs3
cs7 requires cs4
cs8 requires cs5 and cs6
cs9 requires cs7
cs10 requires cs9
cs11 requires cs8
cs12 requires cs3
cs13 requires cs6
cs14 requires cs4 and cs6
cs15 requires cs14

Your task is to determine the minimum number of semesters needed to finish the degree.

(Hint: Represent the courses and their prerequisites as a DAG. Using appropriate data structure to represent the graph).

Please provide:

1. Manually plot the DAG (3%)



2. Explain the algorithm that you are going to implement by the Pseudo-code, and indicate the minimum number of semesters necessary to finish the degree. Indicate the time complexity and space complexity (3%)

The algorithm uses topological sorting with level tracking to find the minimum number of semesters needed. Each course is assigned to the earliest possible semester based on when all its prerequisites are completed.

Pseudo Code

- Input is the G which is the DAG with the courses and the prerequisites
- Initialize: the array of semester levels to 0
 - For each course i in G semester[i] = 0
- Get the topological ordering of courses
 - topoOrder = TopologicalSort(G)
- Calculate semester for each course in the topological order

- For each course c in topoOrder
 - $\text{maxPrereqSemester} = 0$
 - Find each course p in G
 - For each course p in G
 - If p is a prerequisite of c then
 - $\text{maxPrereqSemester} = \max(\text{maxPrereqSemester}, \text{semester}[p])$
 - Assign course to next semester after prerequisite
 - $\text{Semester}[c] = \text{maxPrereqSemester} + 1$
- Find the maximum semester number
 - $\text{maxSemesters} = 0$
 - For each course i in G
 - $\text{maxSemesters} = \max(\text{maxSemesters}, \text{semesters}[i])$
- Return maxSemesters
-

Pseudo for TopologicalSort(G)

- Initialize the indegree array
- Add all the nodes with indegree 0 to the queue
- While the queue is not empty:
 - Remove node from the queue
 - Add to results
 - Decrease indegree of neighbors
 - Add neighbors with indegree 0 to the queue
- Return topological ordering

3. Write and run your program to print out the result for verification (i.e., minimum number of semesters, and print out the running time) (10%)

```
--- Minimum Semesters Calculation ---  
Course scheduling:  
cs1 - semester 1  
cs2 - semester 1  
cs3 - semester 1  
cs4 - semester 2  
cs5 - semester 3  
cs6 - semester 2  
cs7 - semester 3  
cs8 - semester 4  
cs9 - semester 4  
cs10 - semester 5  
cs11 - semester 5  
cs12 - semester 2  
cs13 - semester 3  
cs14 - semester 3  
cs15 - semester 4  
  
MINIMUM SEMESTERS NEEDED: 5  
Running time: 0.024 milliseconds
```

Minimum number of semesters necessary: 5

Semester Schedule:

- Semester 1: cs1, cs2, cs3
- Semester 2: cs4, cs6, cs12
- Semester 3: cs5, cs7, cs13, cs14
- Semester 4: cs8, cs9, cs15
- Semester 5: cs10, cs11

The running time was 0.024 milliseconds

B.2. Linear Programming (10%)

Design and implement a linear programming algorithm to solve the following minimum cost problem

The liquid portion of a diet is to provide at least 300 calories, 36 units of vitamin A, and 90 units of vitamin C daily. A cup of dietary drink X provides 60 calories, 12 units of vitamin A, and 10 units of vitamin C. A cup of dietary drink Y provides 60 calories, 6 units of vitamin A, and 30 units of vitamin C. Now, suppose that dietary drink X costs \$0.12 per cup and drink Y costs \$0.15 per cup. How many cups of each drink should be consumed each day to minimize the cost and still meet the stated daily requirements?

Print out the minimum cost, and the number cups of drink X and number of cups of drink Y;
Print out the running time.

See the project code. These are the results:

- === OPTIMAL SOLUTION ===
- Number of cups of drink X: 3
- Number of cups of drink Y: 2
- Minimum cost: \$0.66
-
- Nutrient totals provided:
- Calories: 300 (required: ≥ 300)
- Vitamin A: 48 (required: ≥ 36)
- Vitamin C: 90 (required: ≥ 90)
- Running time: 0.011 milliseconds

B.3. [Optional Extra point 10%]

To answer the Question #9 (Part A), implement the Simplex Algorithm and display the results by your program, and print out the running time.

```

--- Iteration 3 ---
Entering variable: x3
Ratio test:
  Row 0: 8/5 = 1.6
  Row 2: 9/4 = 2.25
  Row 3: 5/2 = 2.5
Leaving variable: s1
Pivot element: 5
Making pivot element 1 by dividing row 0 by 5
Making row 1 element 0 using factor -3
Making row 2 element 0 using factor 4
Making row 3 element 0 using factor 2
Making row 4 element 0 using factor -5

After pivot operations:
      x1      x2      x3      s1      s2      s3      s4      RHS
-----
s1 0      0      1      0.2     -0.8     0      0.2     1.6
s2 0      1      0      0.6      2.6     0     -1.4     20.8
s3 0      0      0     -0.8      0.2     1      0.2     2.6
s4 1      0      0     -0.4     -0.4     0      0.6     1.8
P  0      0      0      1      6      0      1     268
-----

=== OPTIMAL SOLUTION ===
x1 = 1.8
x2 = 20.8
x3 = 1.6
Maximum P = 268

Running time: 0.137 milliseconds
=====

```

CS 375 Assignment 4 - Part B.3 Simplex Algorithm for Question #9

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Solving using Simplex Method...

Problem: Maximize $P = 20x_1 + 10x_2 + 15x_3$

Initial table:

	x1	x2	x3	s1	s2	s3	s4	RHS
s1	3	2	5	1	0	0	0	55
s2	2	1	1	0	1	0	0	26
s3	1	1	3	0	0	1	0	30
s4	5	2	4	0	0	0	1	57
P	-20	-10	-15	0	0	0	0	0

--- Iteration 1 ---

Entering variable: x1

Ratio test:

Row 0: $55/3 = 18.3333$

Row 1: $26/2 = 13$

Row 2: $30/1 = 30$

Row 3: $57/5 = 11.4$

Leaving variable: s4

Pivot element: 5

Making pivot element 1 by dividing row 3 by 5

Making row 0 element 0 using factor 3

Making row 1 element 0 using factor 2

Making row 2 element 0 using factor 1

Making row 4 element 0 using factor -20

After pivot operations:

	x1	x2	x3	s1	s2	s3	s4	RHS
s1	0	0.8	2.6	1	0	0	-0.6	20.8
s2	0	0.2	-0.6	0	1	0	-0.4	3.2
s3	0	0.6	2.2	0	0	1	-0.2	18.6
s4	1	0.4	0.8	0	0	0	0.2	11.4
P	0	-2	1	0	0	0	4	228

--- Iteration 2 ---

Entering variable: x2

Ratio test:

Row 0: $20.8/0.8 = 26$

Row 1: $3.2/0.2 = 16$

Row 2: $18.6/0.6 = 31$

Row 3: $11.4/0.4 = 28.5$

Leaving variable: s2

Pivot element: 0.2

Making pivot element 1 by dividing row 1 by 0.2

Making row 0 element 0 using factor 0.8

Making row 2 element 0 using factor 0.6

Making row 3 element 0 using factor 0.4

Making row 4 element 0 using factor -2

After pivot operations:

	x1	x2	x3	s1	s2	s3	s4	RHS
s1	0	0	5	1	-4	0	1	8
s2	0	1	-3	0	5	0	-2	16
S3	0	0	4	0	-3	1	1	9

s4	1	0	2	0	-2	0	1	5
P	0	0	-5	0	10	0	1.97215e-31	260

--- Iteration 3 ---

Entering variable: x3

Ratio test:

Row 0: $8/5 = 1.6$

Row 2: $9/4 = 2.25$

Row 3: $5/2 = 2.5$

Leaving variable: s1

Pivot element: 5

Making pivot element 1 by dividing row 0 by 5

Making row 1 element 0 using factor -3

Making row 2 element 0 using factor 4

Making row 3 element 0 using factor 2

Making row 4 element 0 using factor -5

After pivot operations:

	x1	x2	x3	s1	s2	s3	s4	RHS
s1	0	0	1	0.2	-0.8	0	0.2	1.6
S2	0	1	0	0.6	2.6	0	-1.4	20.8
s3	0	0	0	-0.8	0.2	1	0.2	2.6
s4	1	0	0	-0.4	-0.4	0	0.6	1.8
P	0	0	0	1	6	0	1	268

=== OPTIMAL SOLUTION ===

x1 = 1.8

x2 = 20.8

x3 = 1.6

Maximum P = 268

Running time: 0.267 milliseconds

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